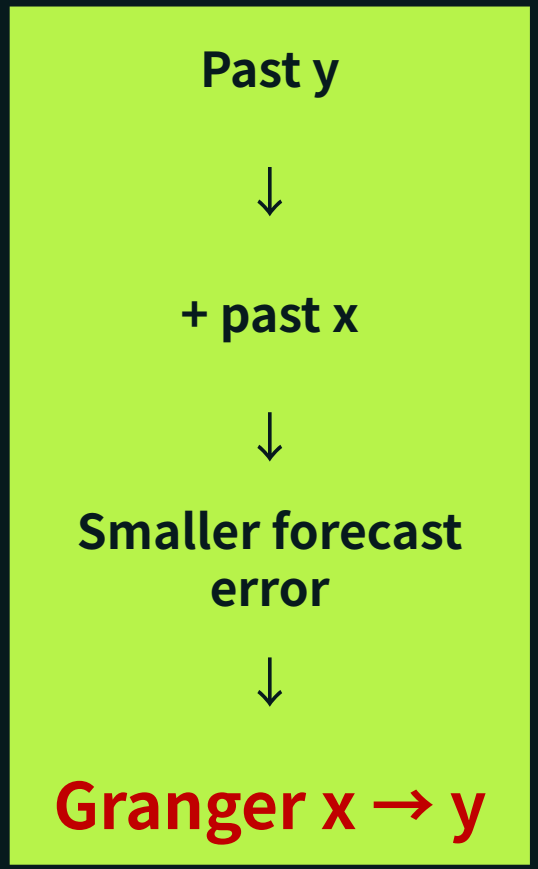
Granger Causality
Predictive Direction

Testing whether the past of x improves the forecast of y



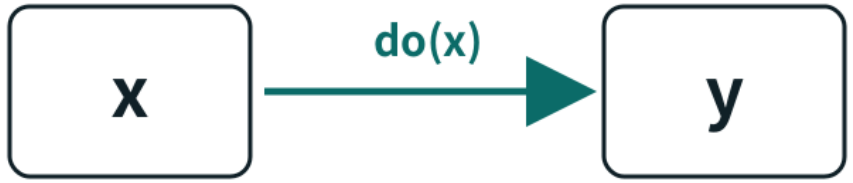
We define, test, and then question Granger causality in four steps

01	**Basic Concepts**	**True vs predictive causality** Definition · time-order intuition · ADL model
02	**Test Procedure**	**Two ADL equations and an F test** Tested coefficients · null · four outcomes
03	**Limits and Pitfalls**	**Linearity · stationarity · omitted variables** Hidden cause · simulation · a news article
04	**Finance Cases and Lab**	**Real lead-lag vs illusion** Futures–spot · term spread · oil · lab

Granger causality tests predictive contribution, not cause

Granger (predictive) causality is distinct from causality about the effect of an intervention

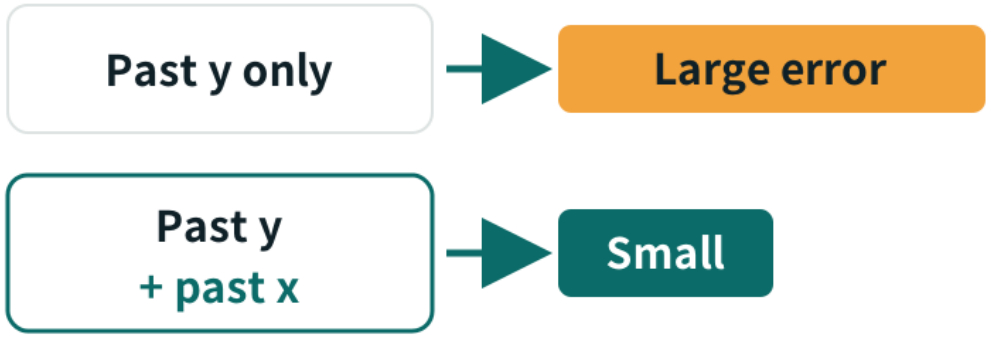
True causality · Intervention



Does changing x directly change y?

Needs experiments or structural models

Predictive causality · Granger



Significantly smaller error → x ⇒ y

If past x improves the forecast of y, x Granger-causes y — predictive help, not a proven cause

The intuition: a cause comes first and carries unique information

Time order (past → present) is used to judge the direction

Intuition 1 · Time order

The cause comes first

- Effects come later
- Past is fixed
- Past values only

Intuition 2 · Unique info

Unique information

- Own past not enough
- Past x adds info
- Extra fit is the key

Practice · Testable

Regression and F test

- Linear OLS estimate
- Test added lags
- Widely used

‘Moves first, adds information’ — the Granger test puts these two conditions into statistics

The base model is an ADL with own lags and the other variable's lags

Autoregressive distributed lag — a intercept, b own lags, c other lags, e current shock

Example with two lags — ADL(2, 2)

$$y_t = a + b_1 y_{t-1} + b_2 y_{t-2} + c_1 x_{t-1} + c_2 x_{t-2} + e_t$$

General form — ADL(p, r): p lags of y, r lags of x

$$y_t = a + b_1 y_{t-1} + \cdots + b_p y_{t-p} + c_1 x_{t-1} + \cdots + c_r x_{t-r} + e_t$$

If the coefficients c on past x are not zero, x adds information that past y does not carry

OLS estimates the coefficients so errors are uncorrelated with the past

All past information is used, leaving no predictable component in the error

Condition · Error properties

Only a pure shock remains

- Error uncorrelated with past
- No predictable part left
- Linear form is the usual choice

Estimation · OLS

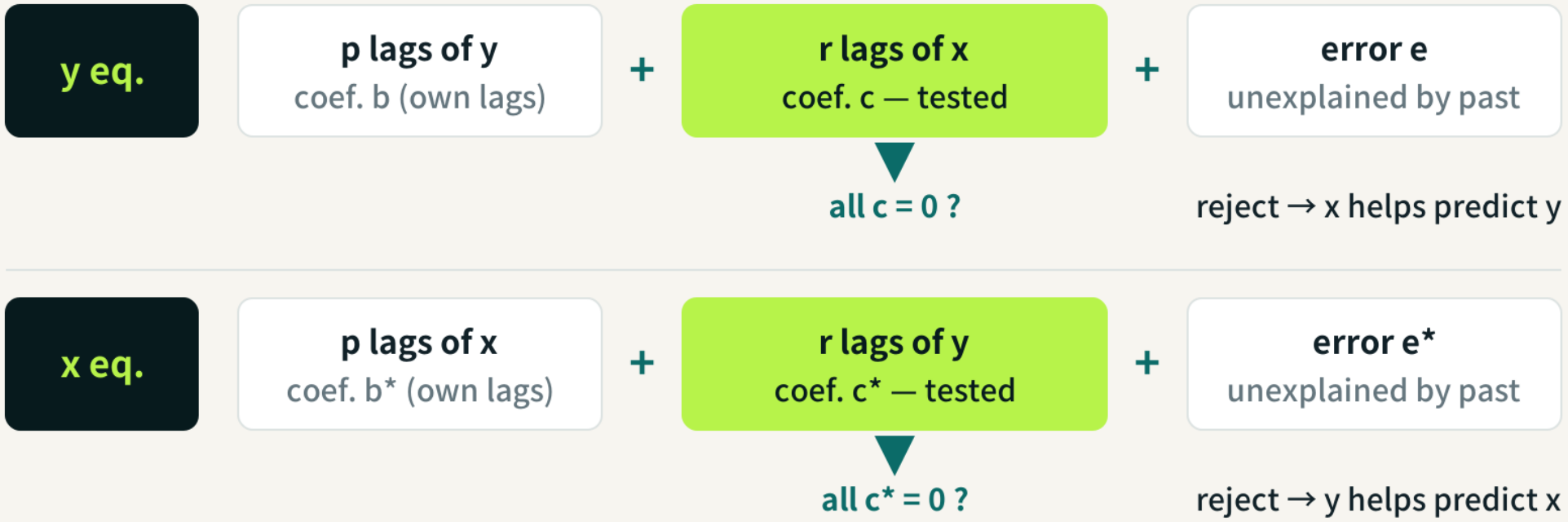
Minimize squared errors

- OLS equation by equation
- Lose the first max-lag obs.
- In effect a bivariate VAR

With a maximum lag of 4, the first 4 observations lack lagged values and drop out

One equation per direction; only the other variable's lags are tested

Lime boxes are tested — c in the y equation checks x→y, c* in the x equation checks y→x



Own lags (b) are controls. What matters is only what the other past adds (c, c*)

An ADL equation of the same form is estimated for y and for x

An asterisk (*) marks the reverse-direction equation — lag lengths may differ between the two

y equation — does past x help predict y? (coefficients c)

$$y_t = a + b_1 y_{t-1} + \cdots + b_p y_{t-p} + c_1 x_{t-1} + \cdots + c_r x_{t-r} + e_t$$

x equation — does past y help predict x? (coefficients c*)

$$x_t = a^* + b_1^* x_{t-1} + \cdots + b_p^* x_{t-p} + c_1^* y_{t-1} + \cdots + c_r^* y_{t-r} + e_t^*$$

Rule: if any c is non-zero and all c* are zero, we say ‘x Granger-causes y’

The F test compares residuals with and without the other's lags

R drops the other variable's lags, U keeps them — run the test in each direction

Null for x → y

$$H_0 : \ c_1 = c_2 = \cdots = c_r = 0$$

Null for y → x

$$H_0 : \ c_1^* = c_2^* = \cdots = c_r^* = 0$$

F statistic

$$F = \frac{(RSS_R - RSS_U)/q}{RSS_U/(n-k)}$$

Notation

- RSS: residual sum of squares
- q: tested lags (restrictions)
- n: obs. · k: coefficients in U

The more the other lags shrink the residuals, the larger F — leaning toward 'helps predict'

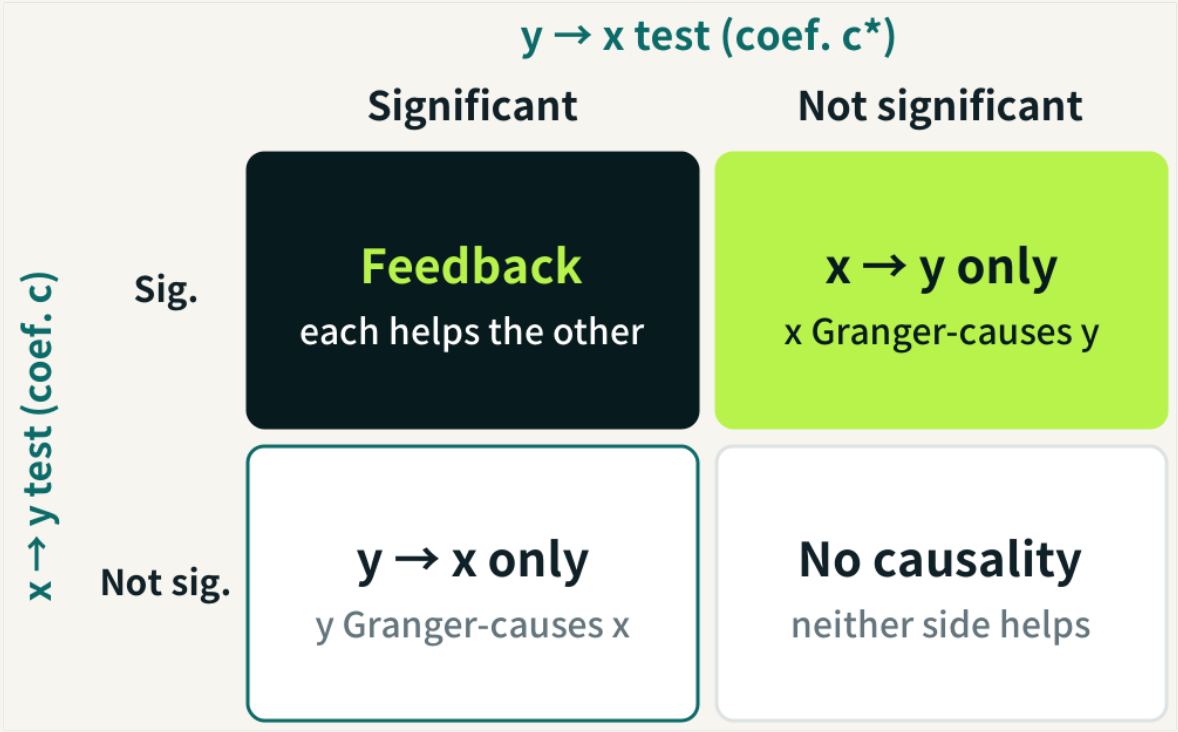
Combining the two directional tests yields four possible conclusions

Rows: significance of the x→y test (c); columns: the y→x test (c*)

Decision rule

- One side: one-way
- Both: feedback
- Test both ways

One direction misses feedback



The Granger test is meaningful only under three assumptions

Linearity · stationarity · dependence on the observed variables

Limit 1

Linearity

- Linear links only
- Nonlinear: hard
- Less-known theory

Limit 2

Stationarity

- Needs stationarity
- I(1): difference
- Coint.: VECM · TY

Limit 3 · Observed variables

Omitted variables

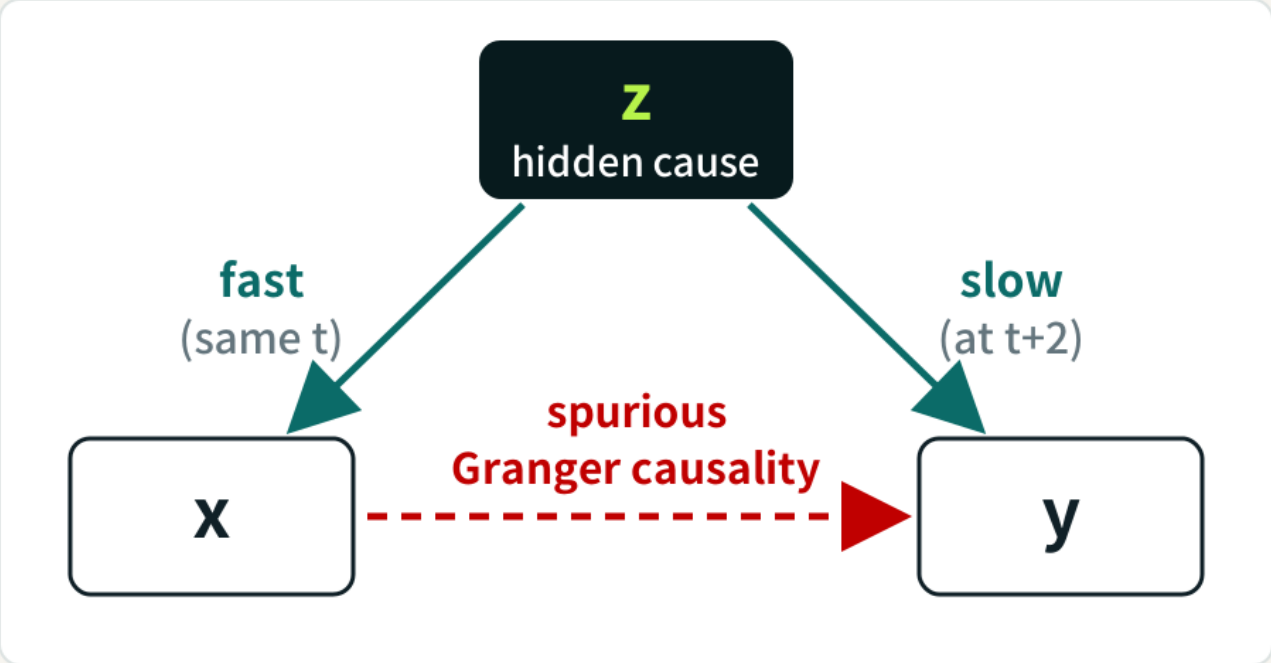
- Included vars only
- Hidden causes
- Most common error

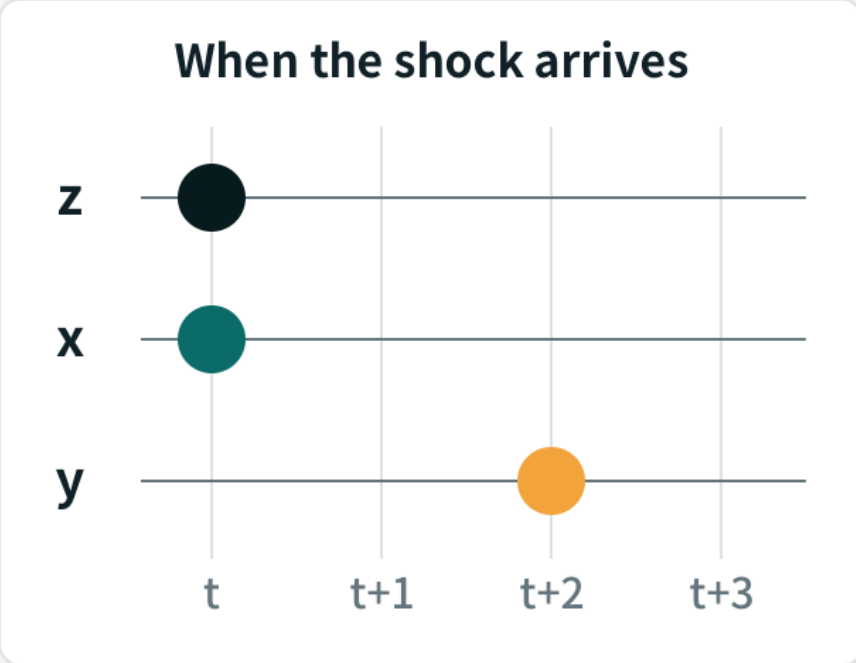
※ TY = Toda–Yamamoto · nonstationary spans may be handled by windowing (Hesse et al., 2003)

The third limit is the most dangerous — the test says nothing about variables you cannot see

A hidden cause with unequal delays creates spurious Granger causality

z hits x quickly and y slowly — the test uses only x and y

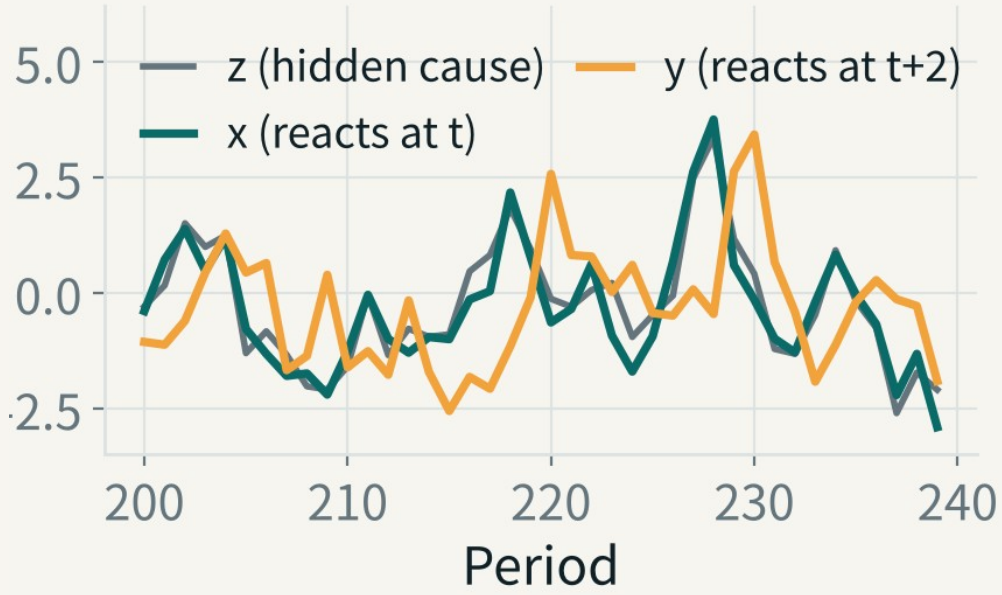


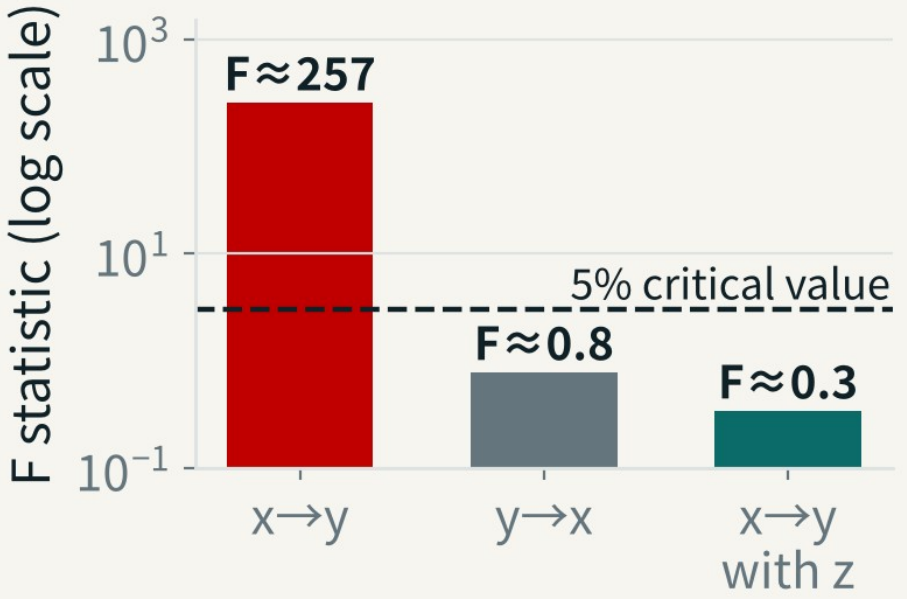


x merely received z's signal first — include z and the x → y causality vanishes

In simulation, too, controlling for z removes x→y Granger causality

z is AR(1) with coefficient 0.6 · x reacts to z at t, y at t+2 · 500 observations, 2 lags





With x and y only F≈257; adding z gives F≈0.3 — the result depends on the variable set

※ Simulated data (seed 42) — 5% critical value F≈3.0

Lag length, frequency, and expectations can change the conclusion

Even with the same data, Granger test results vary with the setup

Caution 1 · Lag choice

Choosing p and r

- Lags change results
- Pick by AIC · BIC
- Vary lags to check

Caution 2 · Frequency

Temporal aggregation

- Annual ≠ daily
- Aggregation hides
- Async. trading

Caution 3 · Expectations

Leads from expectations

- Stocks lead GDP
- Not the cause
- Pricing expectations

Even a significant result lets you say no more than ‘it helps prediction’

The article’s Granger test cannot support its policy conclusion

A Korean newspaper article — real GDP vs Gini coefficient, annual data 1990–2011

The article’s claim

Growth → distribution is strong

- Growth → distribution: strong
- Distribution → growth: weak
- “Growth drives distribution”

Reading it critically

- 22 obs. · low power
- GDP level has a unit root
- Common causes: policy, aging
- Predictive ≠ policy effect

Policy conclusion overreaches

‘Has an effect’ here means predictive contribution only — it says nothing about effect size

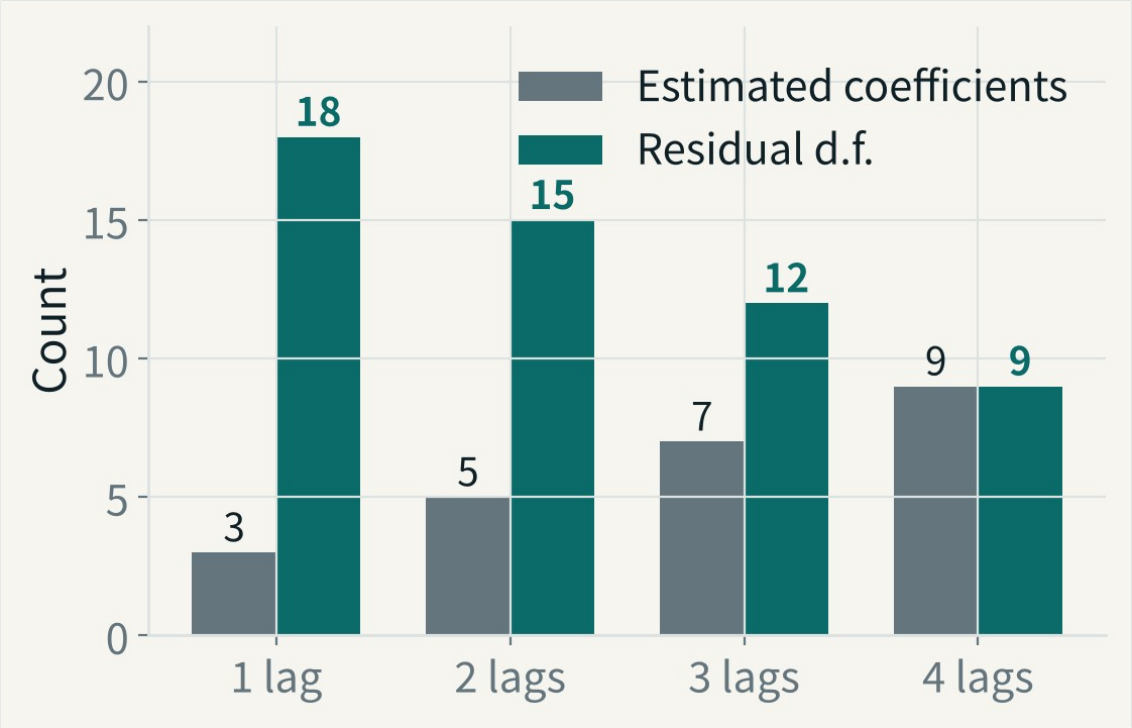
Twenty-two annual observations leave the test badly underpowered

For the y equation — p lags means 1+2p coefficients, and the first p observations drop out

Applied to the article

- 2 coefs per lag
- 4 lags: 9 d.f. left
- Differencing: 21 obs.

No result is reliable



Lead-lag relations appear in markets that absorb information first

Finance cases ① index futures–spot lead-lag ② term spread → business cycle

CASE 1 · Kawaller, Koch & Koch (1987)

Futures lead spot by minutes

- S&P 500 futures ↔ cash index
- Low-cost futures move first
- Beware: stale prices fake leads

CASE 2 · Estrella & Hardouvelis (1991)

Term spread predicts activity

- 10Y–3M spread → real GDP
- Forecasts growth, recession odds
- Reflects expectations, not cause

The source of the lead is speed of information and expectations, not causation

‘No causality’ in a linear test does not mean ‘no relationship’

Finance cases ③ oil prices → real activity ④ trading volume ↔ returns

CASE 3 · Hamilton (1983)

Oil spikes precede recessions

- Oil price changes → U.S. output
- Asymmetric: declines matter less
- One shared coefficient blurs it

CASE 4 · Hiemstra & Jones (1994)

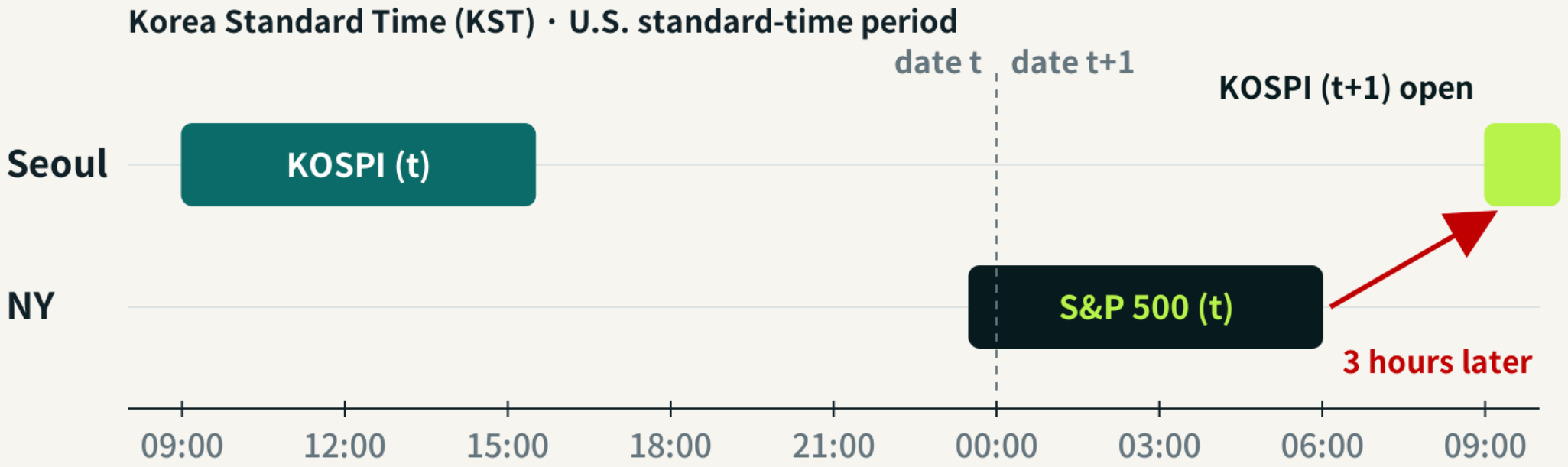
Volume → returns

- Dow Jones volume and returns
- Weak in linear tests
- Found by nonlinear tests

Asymmetric and nonlinear relations slip past the linear F test

Much of U.S.→KOSPI Granger causality comes from trading hours

Finance case ⑤ S&P 500 daily return → next-day KOSPI — Seoul opens after New York closes



Overnight news shows up as S&P(t) → KOSPI(t+1) at lag 1 — realigning alone can flip the result

※ Check time-zone alignment first in cross-country daily tests · async. trading and aggregation also distort

Lab ① checks U.S.→Korea spillover and the date-alignment trap

Claude Code prompt — daily closes of ^GSPC and ^KS11 via yfinance, from 2015-01-01

Download ^GSPC · ^KS11 → **Log returns common days** → **ADF test stationarity** → **F test both ways lags 1–5** → **Alignment same day vs +1**

What to check

Compare two p-value tables

Does the direction flip with alignment?

Output

granger_us_kr.md · .png

Claude Code saves and opens them for you

Students paste the prompt and check results — Claude Code does the rest, even fixing errors

Lab ② tests the term spread's predictive power across subperiods

Claude Code prompt — FRED T10Y3M (monthly avg.) and INDPRO (industrial production), from 1985

Download T10Y3M · INDPRO → **Transform YoY growth** → **Stationarity ADF · KPSS** → **BIC lags both directions** → **Split sample pre/post 2008**

What to check

Does it survive the GFC?

Is the result stable across periods?

Output

granger_term_spread.md

3-line summary + why not a 'cause'

The summary must also state why we cannot say the spread causes the business cycle

Granger causality tests predictive contribution.

Suspect hidden causes, stationarity, time zones first.

Conclude no more than ‘it helps prediction’.

Test both directions,
and report the variable set used.